

Carry-Over Lottery Allocation: Practical Incentive-Compatible Drafts

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

Introduction

For decades, the NBA has navigated the tradeoff between helping weak teams via the draft and the tanking that ensues. Carry-Over Lottery Allocation (COLA) solves the dilemma. COLA redefines weakness, using multi-season playoff results instead of a single season's record. Thus, tanking need not be rewarded. Draft priority carries over between seasons, being reduced when a team has playoff success or makes a high draft pick. Over time, more help is directed to weaker teams.

The COLA framework allows flexibility in how draft priority changes. Its definition includes the NBA's own new 3-2-1 lottery, which has a minimal carry-over exclusion and no reward for losing.

We prove that, under reasonable assumptions, COLA can be incentive-compatible while helping the weakest team, circumventing Banchio and Munro's SSAC 2020 impossibility result for record-based lotteries. Only one assumption requires justification: teams prefer winning games over manipulating the pool of lottery teams. Others are standard assumptions.

Methods

Our incentive-compatibility proof is routine; the novelty is the reframe that makes it possible.

To discuss tradeoffs, we identify tunable dials within the COLA framework:

- How does a team's draft priority increase?
- How does the priority decrease?
- How does priority map to lottery odds and draft positions?

ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move of realistic fictional leagues, lets us compare different draft mechanisms on these criteria:

- **Targeted help.** To what extent do the highest priority teams receive draft help?
- **Long-term parity.** Can all teams be competitive, as measured by long playoff droughts?
- **Simplicity.** Is the mechanism easy to explain?
- **Incentive compatibility.** Does the mechanism reward losing? Is it still anti-tanking if the model's assumptions fail?

Results

We ran nine mechanisms on the same 30 simulated leagues for 25 seasons each. We compare mechanisms by paired differences with 95% confidence intervals.

Simple COLA matches or beats 3-2-1 on simplicity, incentives, and targeted help. On simplicity, it orders teams by most years without a playoff series win or a top-3 draft pick, with the tiebreaker being most wins this season. No lottery is used. Unlike 3-2-1, it helps the weakest teams, where weakness is measured by drought length. Unlike the old NBA lottery, it is incentive-compatible. The improvement in help is statistically significant. On parity, 3-2-1 keeps a slight edge. Where parity matters more, we propose Full-lottery COLA, which draws all 14 lottery picks in proportion to draft priority. It matches 3-2-1 on parity while still giving more help.

Mechanism	Reward for losing <i>draft places gained, 5th-worst to last</i>	Longest stretch, no series win or top-3 pick <i>seasons, any team</i>	Longest drought between series wins <i>seasons, any team, of 25</i>	Average team's longest playoff drought <i>seasons</i>
Old NBA lottery (2019 to 2026)	1.32	16.4	23.3	6.01
New NBA lottery, 3-2-1 (from 2027)	losing costs picks	16.3	24.4	6.03
Simple COLA	losing costs picks	5.5	24.3	6.37
Full-lottery COLA	0	11.9	23.8	5.96

Table 1. Our two COLA proposals and both NBA lotteries on the same 30 simulated leagues of 25 seasons each (means over leagues). The full paper reports all nine mechanisms we ran.

Conclusion

COLA can end tanking incentives while giving the most help to the weakest teams, which record-based lotteries cannot do. The NBA's 3-2-1 lottery, an edge case of the framework, ends tanking incentives but gives almost no help to the weakest teams. Stronger carry-over delivers far more targeted help, and COLA's dials let the league trade off targeted help, parity, simplicity, and incentive compatibility.